

AWS Academy: Cloud Architecting - Capstone Project - Notepad

Ryan Verrot

Pikes Peak State College

CNG 2042 1N1: Cloud Computing

Professor [REDACTED]

2026-05-08 23:59

Environment Documentation

Hostname:	IP Address:	OS/Version:	Users:	Comments:
Ryan-Legion	[REDACTED]	Windows 11 Pro 25H2	[REDACTED]	Personal Windows 11 Pro laptop/workstation utilized to access and complete the AWS Academy: Cloud Architecting - Capstone Project.

Steps Taken

Date & Time (UTC):	Source, Description, and Action Taken:
2026-05-05 21:44:49	Source: Ryan-Legion; AWS Management Console; VPC Dashboard Description: Deployed, accessed, and documented the lab environment required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: 1. From Ryan-Legion, navigated to the AWS Academy: Cloud Architecting - Capstone Project page located at https://awsacademy.instructure.com/courses/154207/assignments/1822927?module_item_id=15009401. 2. From the Capstone Project page, selected the 'Start Lab' button in the upper-right of the screen and waited for the 'AWS' button status indicator to turn green before selecting it to navigate to the AWS Management Console for the Capstone Project. 3. From the AWS Management Console, typed "VPC" in the search bar located at the top of the screen and selected the 'VPC' hyperlink to navigate to the VPC dashboard.

	4. From the VPC dashboard, selected 'Your VPCs' from the left-navigation pane before selecting "Project VPC" on the Your VPCs page and documenting the Resource Map from the resulting screen.
2026-05-05 21:54:44	Source: Ryan-Legion; Description: Created and configured the Amazon RDS MySQL database required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: 1. From the AWS Management Console, typed "RDS" in the search bar located at the top of the screen and selected the 'RDS' hyperlink from the resulting search dropdown menu. 2. From the AWS Aurora and RDS Dashboard, selected 'Databases' from the left navigation pane to navigate to the Databases dashboard before selecting the orange 'Create Database' button from the upper-right and choosing 'Full Configuration' from the resulting dropdown. 3. From the Create Database page, configured an RDS Database with the following settings before selecting the 'Create Database' button located at the bottom of the screen: a. Engine Type: MySQL b. Templates: Dev/Test c. Availability and Durability: Single-AZ DB instance deployment d. DB Instance Identifier: countries-db e. Master Username: admin f. Credentials Management: Managed in AWS Secrets Manager g. DB Instance Class: Burstable Classes h. Instance Type: db.t3.micro i. Allocated Storage: 20 GiB j. Storage Autoscaling: Disabled k. Virtual Private Cloud (VPC): Project VPC l. Existing VPC Security Groups: ExampleDB-SG m. Initial Database Name: countries n. Enhanced Monitoring: Disabled i. All other configuration settings were kept at their default values.
2026-05-05 23:18:33	Source: Ryan-Legion; Description: Created and configured the Target Group for the Application Load Balancer required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken:

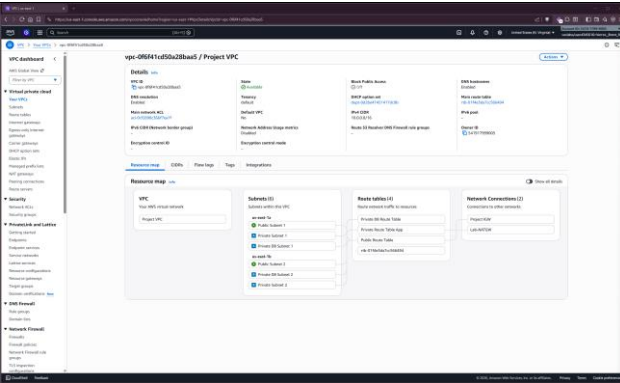
	1. From the AWS Management Console, typed “EC2” in the search bar located at the top of the page and selected the ‘EC2’ hyperlink from the resulting search dropdown menu to navigate to the EC2 Dashboard. 2. From the Amazon EC2 Dashboard, selected ‘Target Groups’ from the left navigation pane before clicking on the orange ‘Create Target Group’ option from the Target Groups page. 3. From the Create Target Group page, configured the Target Group with the following settings before selecting the ‘Create Target Group’ button: a. Target Type: Instances b. Target Group Name: countries-tg c. VPC: Project VPC i. All other configuration settings were kept at their default values.
2026-05-05 23:30:32	Source: Ryan-Legion; Description: Configured and created the Application Load Balancer required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: 1. From the EC2 Dashboard, selected ‘Load Balancers’ from the left navigation pane and clicked on the orange ‘Create Load Balancer’ button in the upper-right and choosing ‘Application Load Balancer’ from the resulting dropdown menu. 2. From the Create Application Load Balancer page, configured the Application Load Balancer with the following settings before selecting the ‘Create Application Load Balancer’ button: a. Load Balancer Name: countries-alb b. VPC: Project VPC c. Availability Zones and Subnets: us-east-1a (Public Subnet 1); us-east-1b (Public Subnet 2) d. Security Groups: ALBSG e. Forward to Target Group: countries-tg i. All other configuration settings were kept set to their default values, with the exception of the removal of the default security group.
2026-05-05 23:41:38	Source: Ryan-Legion; Description: Configured and created the Auto Scaling Group required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: 1. From the EC2 Dashboard, selected ‘Auto Scaling Group’ from the left navigation pane and clicked on the orange

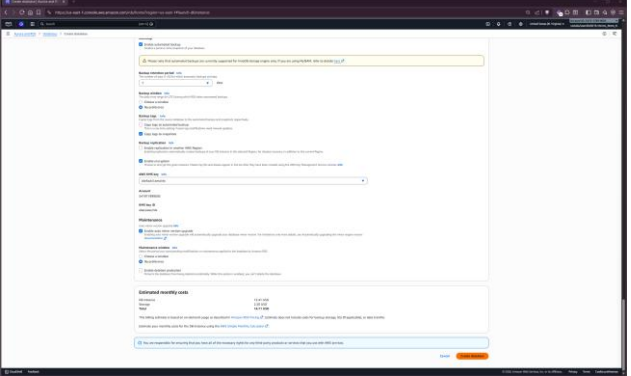
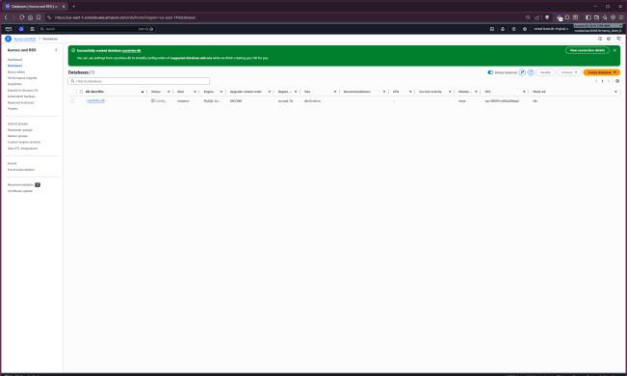
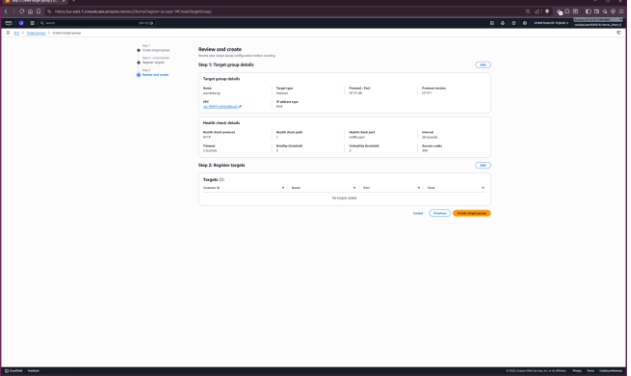
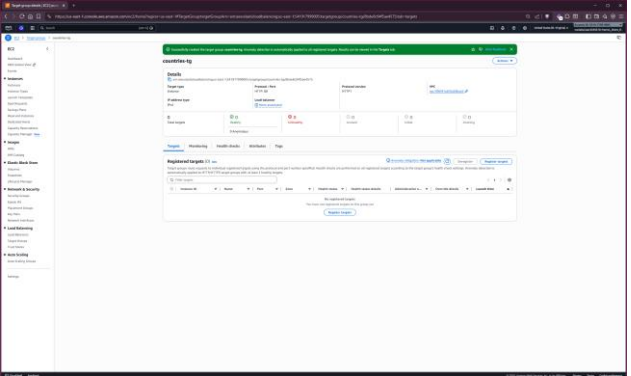
	‘Create Auto Scaling Group’ button from the upper-right of the page. - From the Choose Launch Template page, configured the Auto Scaling Group with the following settings before selecting the orange ‘Create’ button from the bottom-right of the page: - Auto Scaling Group Name: countries-asg - Launch Template: Project-LT - Version: Latest (1) - VPC: Project VPC - Availability Zones and Subnets: Private Subnet 1; Private Subnet 2 - Select Load Balancing Options: Attach to an Existing Load Balancer - Existing Load Balancer Target Groups: countries-tg - Additional Health Check Types: Turn on Elastic Load Balancing Health Checks - Desired Capacity: 2 - Minimum Desired Capacity: 2 - Maximum Desired Capacity: 4 - Choose Whether to Use a Target Tracking Policy: Target Tracking Scaling Policy - All other configuration settings were kept set to their default values. - From the EC2 Auto Scaling Groups page, waited until the Instance Health on the countries-asg Auto Scaling Group passed 2/2 Healthy.
2026-05-06 00:13:57	Source: Ryan-Legion; Description: Connected to an instance and imported the SQL data required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: - From the EC2 Dashboard, selected ‘Instances’ from the left navigation pane before choosing the “ExampleAPP” instance running in us-east-1a with Instance ID: i-0f863a148a10b8879 and clicking the ‘Connect’ button in the upper-right. - From the resulting Connect page, selected the ‘SSM Session Manager’ tab before clicking on the orange ‘Connect’ button in the bottom-right. - From the resulting SSM Systems Manager terminal, utilized the command ‘sudo su - ec2-user’ to switch to the ec2-user account before using the command ‘wget

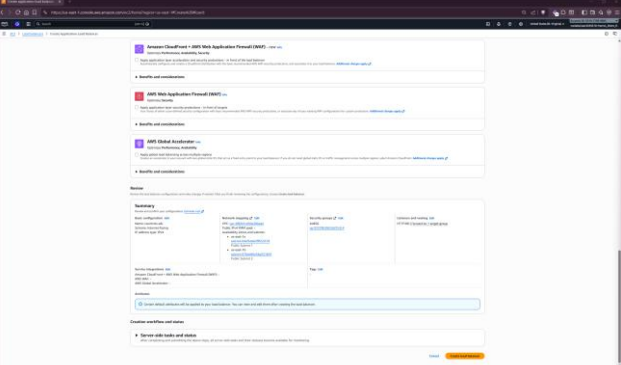
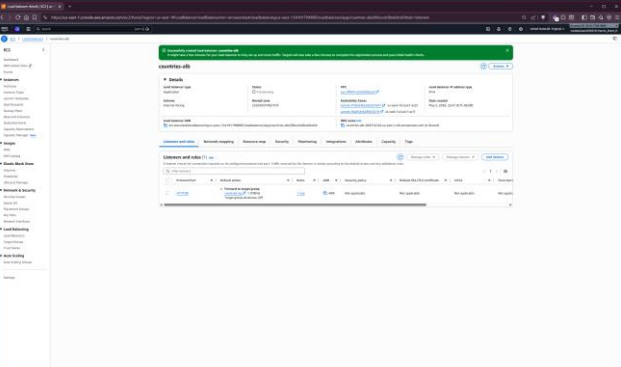
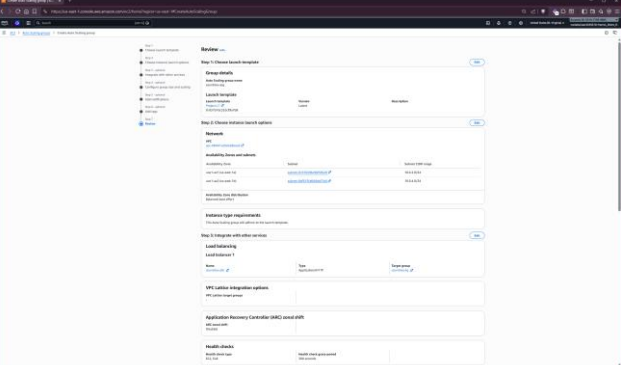
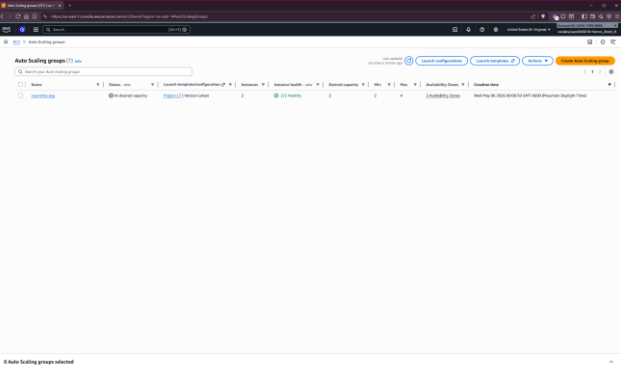
	lab-Capstone-project/s3/Countrydatadump.sql' to download the required Countrydatadump.sql file. - From the SSM Systems Manager terminal, utilized the command 'aws secretsmanager list-secrets --query "SecretList[*].Name" --output text' to identify the name of the secret as rds!db-b987f854-6f9f-42a5-85f9-fd5b9274024c. - From the SSM Systems Manager terminal, utilized the command 'aws secretsmanager get-secret-value --secret-id 'rds!db-b987f854-6f9f-42a5-85f9-fd5b9274024c' --query SecretString --output text' to identify the username and password in JSON output as {"username":"admin","password":"dwrN)8Hln)Lrzv4Ns07HbSdiT:~Q"}. - From the SSM Systems Manager terminal, utilized the command 'mysql -h countries-db.cjssw6k0pu7e.us-east-1.rds.amazonaws.com -u admin -p countries < /home/ec2-user/Countrydatadump.sql' and input the previously identified admin password to import the data from the Countrydatadump.sql file to the countries-db database. - From the SSM Systems Manager terminal, utilized the command 'mysql -h countries-db.cjssw6k0pu7e.us-east-1.rds.amazonaws.com -u admin -p countries' and entered the password dwrN)8Hln)Lrzv4Ns07HbSdiT:~Q to access the mysql database. - From the MySQL command-line, utilized the command 'SHOW TABLES;' to list the tables where I was able to observe the countrydata_final table. - From the MySQL command-line, utilized the command 'SELECT * FROM countrydata_final LIMIT 5;' for additional confirmation that data was present on database before utilizing the 'exit' command to leave the MySQL command-line terminal.
2026-05-06 01:06:29	Source: Ryan-Legion; Description: Tested the Application Load Balancer URL and Web Application Query page to ensure functionality required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: - From the EC2 Dashboard, selected 'Load Balancers' from the left navigation pane before choosing the countries-alb load balancer where I was able to observe that the DNS name was countries-alb-269316742.us-east-1.elb.amazonaws.com. - From a new browser tab, navigated to the web application located at http://countries-alb-

	269316742.us-east-1.elb.amazonaws.com/index.php and clicked on the 'Query' hyperlink in the upper-left of the page. 3. From the resulting Query page, selected 'Mobile Phones' and 'Population' from the dropdown menu, clicking on submit each time to test functionality of the application, which resulted in me being able to observe output from each query.
2026-05-06 01:47:34	Source: Ryan Legion; Description: Submitted the AWS Academy: Cloud Architecting - Capstone Project for grading. Action Taken: 1. From the Capstone Project page, selected the 'Submit' button from the upper-right of the screen. The grade returned as a 35/35 (100%), bringing the AWS Academy: Cloud Architecting - Capstone Project to a conclusion. Additionally, the Grades page in the AWS Academy courseware split the Capstone Project into two entries with one indicating 35/35 and another indicating 56/56.

Evidence Log

Date & Time (UTC):	Action Taken:	Evidence:
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2026-05-06
00:13:57

Connected to
an instance
and imported
the SQL data
required to
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The screenshot shows a terminal window with a dark background. It displays a series of SQL commands and their output. The commands include creating a table named 'users' with columns for id, name, email, password, and role. The output shows the table being created successfully. Below this, there is a list of users with their respective details, including id, name, email, password, and role. The list is formatted as a table with columns for each attribute.

```
CREATE TABLE users (
  id INT(11) UNSIGNED NOT NULL AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(255) NOT NULL,
  email VARCHAR(255) NOT NULL,
  password VARCHAR(255) NOT NULL,
  role VARCHAR(255) NOT NULL
) ENGINE=InnoDB;

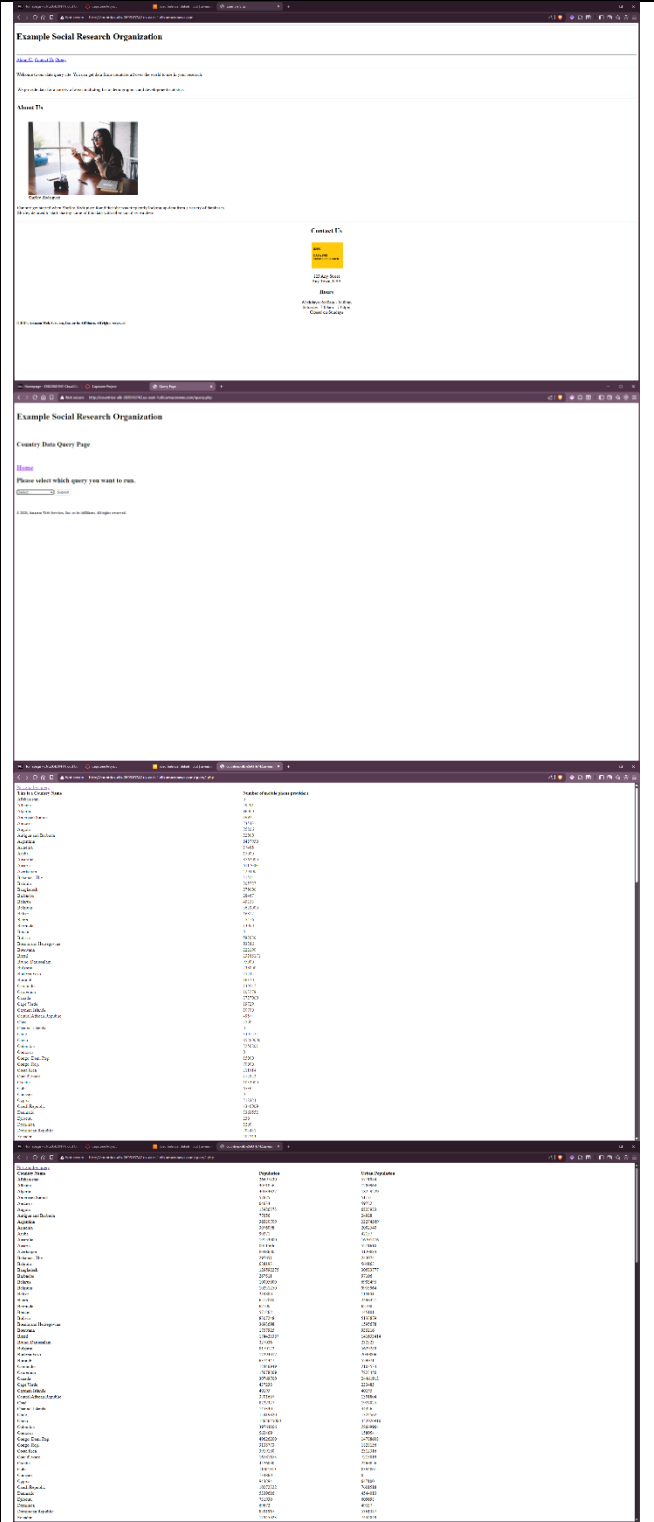
INSERT INTO users (name, email, password, role) VALUES
('John Doe', 'john.doe@example.com', 'password123', 'user'),
('Jane Smith', 'jane.smith@example.com', 'password456', 'user'),
('Admin User', 'admin@example.com', 'admin123', 'admin');

SELECT * FROM users;
```

id	name	email	password	role
1	John Doe	john.doe@example.com	password123	user
2	Jane Smith	jane.smith@example.com	password456	user
3	Admin User	admin@example.com	admin123	admin

2026-05-06
01:06:29

Tested the Application Load Balancer URL and Web Application Query page to ensure functionality required to complete the AWS Academy: Cloud Architecting - Capstone Project.



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01:47:34

Submitted the
AWS Academy:
Cloud
Architecting -
Capstone
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grading.

